

TIMOTHY GIBBENS

AD Architect & Engineer

AD DS/Azure/Entra ID/IaaS

Security, Automation & Tools

PROFILE

Experienced Directory Services architect and engineer with extensive expertise in Azure Entra ID, Active Directory & Microsoft Windows Server technologies. Skilled in hands-on project work, team collaboration, and providing comprehensive solutions for Identity Management lifecycle across cloud and on-premises infrastructures. Proficient in integrating Microsoft Technologies via native and third-party software, as well as scripted and automated solutions. Committed to service excellence, ensuring security, adherence to policies, and industry best practices while exceeding customer expectations.

CONTACT

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EMPLOYMENT SUMMARY

PepsiCo – Systems Engineer & Architect

2023 – 2025

Architect, engineer and support a highly complex, enterprise-level Active Directory in a hybrid on-premises cloud-hosted environment. Manage enterprise identity cloud directories, including Microsoft AD, Entra ID & IAM Infrastructure & Resources. Develop automated solutions, detailed architecture, standards, design and implementation documentation, and continuous improvement action plans.

Kyndryl – AD/ADFS SME

2022 – 2023

Support and stabilize the client's AD & ADFS infrastructure, including the integration of IT applications, services and tools, AD configuration cleanup and remediation, and patching/backup/recovery.

HCL – Consultant IT

2014 – 2022

Azure/M365 Tenant and Global Admin role, integrating on premises AD and resources via AADC, ADFS & IaaS. Directory Services engineer designing, implementing, and supporting corporate infrastructure, disaster recovery and security compliance. Support and facilitate the Transformation and Migration Project to a fully managed services model.

PepsiCo – Senior IT Specialist

1998 – 2014

Engineer and support international migration and divestiture projects. Design, implement and support AD corporate infrastructures. Provide backline AD support and training for all engineered solutions.

EDUCATION

LAN Engineer & Administrator

1996 – 1997

Durham Business and Computer College

Bachelor of Arts Degree

1990 – 1993

University of Toronto

SKILLS

- Windows Server (AD; LDAP; LDS; GPO; DFS; ADFS; DNS; WINS)
- Azure-Entra ID-IaaS-PaaS-SaaS-M365 technologies
- Backup/Recovery: Quest RMAD; Veritas; Symantec, EMC
- Security and Audit: Okta; ChangeAuditor; Splunk; Qualys; Tripwire
- Automation and Scripting: Batch; VBScript; PowerShell; CLI; JSON
- Windows Client technologies

FULL WORK EXPERIENCE - DETAILED

PepsiCo – Systems Engineer & Architect

May 2023 – Sep 2025

Support corporate infrastructure:

Analyze, design, and support a highly complex, enterprise-level Active Directory service in a hybrid on-premises and cloud-hosted environment.

Architect and engineer enterprise identity cloud directories, including Microsoft AD and Azure Entra ID.

Develop and maintain Conditional Access Policies and dovetail with Azure PIM roles.

Administer and support International AD DNS solutions integrating with global BIND DNS systems.

Participate in or lead troubleshooting and incident resolution of complex high severity incidents.

Provide architectural and support guidance on both AD and Azure cloud trusts.

Develop detailed architecture, standards, design, and implementation documentation.

Analyze the current Directory Services environment to identify technical and operational opportunities and develop continuous improvement action plans.

Leverage organizational SCRUM and agile methods via Azure DevOps to build and implement all projects and out-of-band operations deployments.

Collaborate with the Intune device management team on project initiatives and BAU operations work.

Support M365 (O365) engineering and operations teams' efforts on multiple fronts, including primarily: email, OneDrive, licensing, security, sensitivity labels & forms.

Managed IAM-owned Azure VM virtual machines and other IaaS resources including capacity improvements, automation accounts & runbooks, and Log Analytics workspaces.

Developed multiple automated solutions for streamlined AD & Entra ID administration using PowerShell scripts:

- Stale devices & inactive objects, account terminations & legal hold, DNS record mgmt, Tier 0 PIM enablement, Entra Role and Azure RBAC role assignments, MFA reset, App Secrets & API permissions, M365 licensing, and various object updates and reports.
- Ran Azure CLI Terraform updates against Entra ID RBAC for Cyera, WIZ & others.

Project: SWIFT Security

Conduct an architectural security review of the Windows based PepsiCo SWIFT financial system and create a proposal with recommendations for optimizing security and privileged access to the SWIFT infrastructure.

Project: Greenfield OT Forest

Architect, engineer and build the new greenfield OT AD Forest, Windows server & Azure networking infrastructure, automations, audit/monitoring/alerting tools and processes. Implemented fully engineered tiered WSUS patching for DCs and member servers. Create HLD & LLD design docs and runbooks.

Project: AD Attack Path Analysis

Architectural review and decision of new attack path security tool. Perform full financial analysis with risk impact & tool procurement. Engineer and deploy tool infrastructure. Provide high level guidance to Operations team of remediating indicators of exposure.

Project: Azure China Integration

Architect and integrate China 21 Vianet-managed Azure cloud via Entra Connect Sync leveraging globally managed identities and passwords including integration with Azure AD Domain Services supporting Kerberos and DNS access for IaaS hosted Windows servers.

Project: Azure Cloud Sync Pilot

Perform PoC implementation to sync users and groups into Entra ID to replace Entra Connect Sync

Project: Windows Hello for Business

Review multiple solutions for integration and coordinate alignment & approval between all supporting teams. Implement the preferred option of Cloud Kerberos Trust in the Azure tenant and Active Directory.

Security and compliance:

Engineer and develop continuous improvements to AD GPOs across the forest to enhance both client performance and security best practices across all Windows client/server/application platforms.

Architect and revise privileged access assignments for Azure Purview, Exchange & Compliance Security portals. Scripted automation of IAM integrated processes and reports; account and password notifications; AD & server health monitoring.

Identify and address AD access security violations through tools/monitoring/alerting systems and be an advocate for following security processes and procedures.

Lead and guide AD team on all AD policies and standards.

Project: Azure RBAC & PIM

Architect and engineer IAM PIM solutions for: AD Domain Admin privileged access; Azure Entra ID privileged roles; Azure resource RBAC privileged access.

Project: Inter-cloud PIM

Integrate and secure various cloud platforms for Azure Corporate, Azure China, AWS, GCP.

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Project: Tiered Access

Architect and implement a tiered access model for both AD domain controllers and Windows member servers focusing on the principals of least privilege and elimination of AD attack path escalation.

Project: MS Assessment

Implement Microsoft AD and Entra ID security and performance assessments. Review and prioritize findings based on organizational requirements. Perform all remediation work that enhanced security and improved service performance.

Project: Crowdstrike Assessment

Support implementation and review findings for AD. Make security improvements across all AD security areas with a focus on authentication protocols (Kerberos, NTLM and PKI) and communications (LDAP signed and SSL).

Kyndryl – AD/ADFS SME

Feb 2022 – May 2023

Support corporate infrastructure:

Provide BAU AD backline support for multiple AD forests and NT domains.

Perform exhaustive AD/DC/DNS/DFS/Site/Subnet configuration cleanup and remediation of backlogged items. Perform ADFS updates including implementing the AD password change portal; certificate refresh; service failure automation and notification.

Create/document the AD Architecture, DR, SOP, promotion/demotion/refresh, server inventory & ADFS infrastructure processes.

Manage and true-up AD CMDB CIs and catalog items.

Perform and maintain quarterly AD architecture reporting and publication for IT consumption.

Administer and support AD integrated DNS with global BIND DNS systems.

Work with Storage teams to troubleshoot and migrate legacy SAMBA shares to NAS, and finally, retire all legacy NT Domain Infrastructure.

Project: Cross platform integrations

Integrate AD with applications and IT infrastructure including: Wintel; Hyper-V; storage; SAN; DNS; Citrix; networking; SQL; firewall; clustering (incl Database); hardware/software monitoring toolsets; PKI certificate services.

Project: AD Backup Improvements

Successfully re-configure AD & DC backups via EMC Networker.

Project: DC Patching

Deploy WinRM/WSUS via GPO for automated Wintel patching for all domain member servers via Ansible.

Security and compliance:

Create and manage PowerShell scripted automation of IAM integrated processes and reports; account and password notifications; AD & server health monitoring.

Create AD vault to secure and onboard highly privileged accounts.

Identify and address AD access security violations through tools/monitoring/alerting systems and be an advocate for following security processes and procedures.

Support Qualys vulnerability remediation via scripted processes and GPO.

Lead and guide AD team on all AD policies and standards.

Process AD Service and Identity Management requests and provide guidance and training to infrastructure support teams on AD and integrated processes.

HCL – Consultant IT

July 2014 – Feb 2022

Azure/Office365 Tenant Management and Global Admin Role:

Support Office365 federated cloud solution as a global admin. Support and maintain the AAD (Entra) Connect/FIM sync of hybrid accounts and automated licensing model.

SME and engineer for “ADFS” authentication infrastructure including WAP proxy/application publishing.

Develop, customize, and maintain ADFS integration with multiple IDPs and SPs.

Assist with implementation of IaaS via direct on-premises authentication/extended network.

Project: DC Health

Implement and maintain AAD (Entra) Connect Health monitoring of ADFS and AD systems.

Project: ADFS Upgrade

Manage and complete the ADFS version 3.0 upgrade and load balanced solution. Design and planning of the ADFS 4.0 upgrade.

Design, implement and support corporate infrastructures:

Provide AD support for multiple AD forests, hundreds of domain controllers, and hundreds of thousands of Windows computers/servers and users.

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Directory Services engineer and Enterprise/Schema Admin duties: AD forest & domain infrastructure design, engineering, management, server build, backline support and integration with applications and IT infrastructure including Wintel; VMWare; storage; SAN; DNS; Citrix; networking; Exchange; SQL; firewall; clustering (incl Database); hardware/software monitoring toolsets; PKI certificate services; RSA.

Drive and support AD server refresh to meet BAU and project upgrade requirements.

Perform and maintain quarterly AD architecture reporting and publication for IT consumption including AD metrics and object counts across sectors.

Administer and support International AD DNS and corporate WINS solutions integrating with global BIND DNS systems.

Analyze metrics and engineer improvements in AD services and processes to reduce SLA volume and response times. Support and guide AD team with escalations

Project: Exchange Schema Upgrades

Perform forest/domain prep and installation of Exchange server in AD forest.

Project: AD Backups

Engineer, maintain, support and document AD backup & recovery processes.

Project: AD Disaster Recovery Model

Architect, engineer and support an AD solution for DR across all data centers worldwide. Perform quarterly recovery testing in alignment planned DR exercises.

Project: AD in the DMZ

Develop and deploy isolated AD in the DMZ to support consolidated access and management requirements.

Security and compliance:

Perform AD security compliance and remediation processes by following corporate global security policies.

Identify and address AD access security violations through tools/monitoring/alerting systems and be an advocate for following security processes and procedures.

Creation and publication of documented AD processes and standards which align with global policies. Maintain and update 3-year AD roadmap.

Lead and guide AD team patch management of test and production AD infrastructure through WSUS and IEM.

Evaluate all known vulnerabilities and determine remediation processes and solutions.

Support Qualys & Tripwire vulnerability remediation via scripted processes and GPO

Lead and guide AD team on all AD policies and standards, including deployment of engineered solutions.

Process AD Service and Identity Management requests and provide guidance and training to infrastructure support teams on AD and integrated processes.

Project: IAM AD Provisioning

Configure AD to integrate with IAM security tools for provisioning/de-provisioning user access and support workflow processes with Identity Management & Password Manager synchronization.

Project: CyberArk Implementation

Assist with integration of CyberArk solution to provide secure administrative access to AD, applications, and servers, with frequent password rollover.

Support Transformation and Migration Project:

Develop and engineer strategy and solution to provide AD directory and integrated DNS services to transform and migrate servers throughout enterprise across all divisions to new vendor managed support model.

Coordinate with infrastructure teams using isolated networks to provide and support a cohesive migration solution that executed on the delivery of 3000+ servers to production which ran 24/7/365.

Develop and build out production configuration to support all international servers into consolidated administration model.

Assist and/or engineer and deliver on the migration and retirement of legacy AD domains and servers.

PepsiCo – Senior IT Specialist

Jan 2007 – July 2014

Support International migration and divestiture projects for all of PepsiCo:

Support migration of PI regions into one AD domain and DNS namespace under the Global Directory infrastructure via the establishment of bi-directional, SID filtered domain trusts; the migration of users, computers and groups using Quest Migration Manager and scripted tools; and access to trusted resources.

Support divestiture of PepsiCo business units into or under external partners.

Creation and publication of documented processes.

Design, implement and support PepsiCo corporate infrastructures:

AD forest & domain infrastructure design, engineering, management, support, and integration through Identity management.

Baseline management and engineering of procurement/refresh of AD server hardware and service, and engineer and maintain AD recovery processes.

Perform AD security compliance and remediation processes.

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Perform and maintain automated AD service metric reporting.
Design, maintain and support test and certification AD infrastructure.
Patch management of test and certification AD & mail infrastructure.
Enterprise and Schema Admin responsibilities for PepsiCo.
Support and administer implementation of member server password solutions.
Administer and support International AD DNS and corporate WINS solutions.
Develop and document AD standards and processes.
Engineer improvements and automation of AD services and processes to reduce SLA volume and response times.
Project: Software Distribution
Design, engineer, implement and support the Software distribution solution via ADAM/LDS.
Project: Azure Sync and Federation
Design and engineering of federated PepsiCo Azure/O365 solution via FIM sync and Cloud based authentication "ADFS" infrastructure, including the development, customization, and maintenance of the ADFS 2.0 load balanced solution.

Provide backline Active Directory support for all AD engineered solutions including on call after hours support:

Including training and guidance to Steady State AD team of AD processes, standards, and deployment of engineered solutions.
Process AD Service and Identity management requests and provide guidance and training to infrastructure support teams on AD and integrated processes.

Pepsi Bottling Group – Senior Infrastructure Specialist

May 1998 – Dec 2006

International project to align cross-border infrastructure engineering efforts to reduce administration, support and IT costs by providing a simplified and unified North American solution:

Developing a migration strategy for 400+ NDS/Netware/eDirectory servers to AD/Windows file and print including the engineering, testing and implementation of the solution.
Unifying application deployment, desktop, and patch management solutions.
Integrating antivirus protection infrastructures.

Design, implement and support Canadian corporate infrastructures:

NT-AD domain/NetIQ eDirectory design, management, support and including integration through Novell Identity management and dirXML technologies.
Password synchronization configuration and management via NIM & dirXML.
Creation and administration of workstation policies and applications.
Provision and implementation of corporate hardware/software standards and solutions for both workstations and servers.
File, print and application server testing, configuration, implementation, and management.
Capacity planning, upgrading and maintenance of file and application servers.
Centralized configuration and management of DNS/DHCP and WINS.
Server and workstation patch management and reporting.
Antivirus infrastructure and client virus definition management and updating.

Corporate Intranet Web master:

Publishing of corporate information and documentation.
Administration of supporting SQL server database.

Provide third-level support and resolutions for user and field-related issues

Jet Composites Inc – Network Administrator

May 1996 – Apr 1998

Infrastructure Configuration and Management:

Hardware/software configuration, maintenance and deployment of workstations and servers.
Server backup installation, configuration, and restoration.
Network administration of NT and NDS directories, user accounts, filesystems, and printing.
Installation, configuration, deployment of corporate e-mail solution GroupWise across WAN and the Internet.

Application Deployments:

Installation and configuration of server-based fax software utilizing modem-sharing.
Installation and configuration of centrally controlled time-clock hardware across sites.
Installation, configuration and management of IIS web server and firewall for corporate website.

Training & Support:

Corporate user training of network software and standards.
Troubleshooting and repair of all network and client-related issues.

CERTIFICATIONS

AWS Certified Cloud Practitioner	2023
Microsoft Certified: Azure Fundamentals	2023
Microsoft Certified Professional: Administering Microsoft SQL Server 7.0	1999
MCSE - Microsoft Certified Systems Engineer (NT Workstation; NT Server; NT Enterprise Server; Internetworking with Microsoft TCP/IP)	1997-1998
Novell Certified NetWare & eDirectory Engineer 4.1, 5.0 & 6.0 (Administration; Advanced Administration; Service and Support; Design and Implementation; Network Technologies; Installation and Configuration; TCP/IP; 5.0 & 6.0 Updates)	1997-2000

EDUCATION

Certification training	(IN PROGRESS)
Identity and Access Administrator Associate	
Azure Administrator Associate	
Azure Solutions Architect Expert	
Microsoft Training	Jun 2006
Administering and Troubleshooting Active Directory	
Durham Business and Computer College	Nov 1996 – May 1997
LAN Administrator and LAN Engineer Certificate	
Supporting Microsoft NT 4.0 Workstation	
Supporting Microsoft NT 4.0 Server	
University of Toronto	Sep 1990 – May 1993
Bachelor of Arts Degree	